Road Event Detector Formulations

road_events documentation

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Contents

1 Scope

This document records the math and streaming algorithms implemented by the `road_events` crate. The crate detects road events from vehicle-frame motion signals produced by the simulation and visualization pipeline. The detectors are intentionally small-state and embedded-friendly: they process one sample at a time, use bounded memory, and avoid FFTs, dynamic allocation, and offline window buffering.

The detectors are not part of the EKF state estimator. They consume already estimated or referenced motion channels such as vehicle speed, vehicle pitch, gravity-compensated vertical acceleration, and yaw rate. The current event set is:

- speed bumps,
- sustained uphill and downhill intervals,
- reverse-driving intervals,
- harsh acceleration,
- harsh braking,
- harsh cornering,
- road-surface roughness,
- streaming trip statistics such as distance, duration, speed extrema, elevation gain/loss, and event rates.

2 Signals and Units

Symbol	Unit	Meaning
t_k	s	Sample timestamp.
v_x	m/s	Vehicle-frame forward velocity. Positive is forward; negative is reverse.
$s = \lvert v_x \rvert$	m/s	Speed magnitude used by threshold logic.
θ	deg	Vehicle pitch angle. Positive pitch is uphill in the current detector convention.
a_z	m/s^2	Gravity-compensated vehicle-frame vertical acceleration.
$a_{z,\mathrm{bp}}$	m/s^2	Band-passed vertical acceleration used for road roughness.
ω_z	rad/s	Vehicle yaw rate.

The speed-bump detector uses pitch and vertical acceleration. Road roughness uses gravity-compensated vertical acceleration and speed. Hill detection uses pitch and speed. Reverse and harsh longitudinal detection use forward velocity. Harsh cornering uses speed and yaw rate. Trip statistics consume the same vehicle-motion stream and update running totals without storing history.

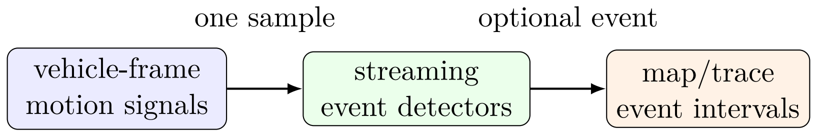

Figure 1: All event detectors run as streaming post-processors over vehicle motion signals.

3 Shared Streaming Primitives

3.1 Timestamp Handling

The implementation clamps per-sample time deltas to keep bad or repeated timestamps from creating unbounded filter updates:

$$\Delta t_k = \mathrm{clamp}(t_k - t_{k-1}, 0, 0.2). \tag{1}$$

For high-pass and derivative paths that require a strictly positive time step, the lower bound is 10^{-3} or 10^{-4} seconds depending on the path.

3.2 First-Order High-Pass Filter

The speed-bump detector removes road grade and slow body motion with the standard discrete first-order high-pass filter:

$$\tau = \frac{1}{2\pi f_c}, \tag{2}$$

$$\alpha_k = \frac{\tau}{\tau + \Delta t_k}, \tag{3}$$

$$y_k = \alpha_k (y_{k-1} + u_k - u_{k-1}). \tag{4}$$

This is applied independently to vehicle pitch and gravity-compensated vertical acceleration.

3.3 Exponential Moving Average

The crate uses an exponential moving average (EMA) for adaptive noise floors and smoothed velocity derivatives:

$$\alpha_k = \frac{\Delta t_k}{\max(\tau, \Delta t_k) + \Delta t_k}, \tag{5}$$

$$\bar{x}_k = (1 - \alpha_k)\bar{x}_{k-1} + \alpha_k x_k. \tag{6}$$

For noise floors, the input is the absolute high-pass value:

$$\sigma_k = \mathrm{EMA}(\sigma_{k-1}, |y_k|, \tau, \Delta t_k). \tag{7}$$

The EMA is a low-memory substitute for a sliding RMS window. It provides slow adaptation to background vibration without storing historical samples.

3.4 Distance-Domain EMA

Road roughness should compare road surface quality over distance rather than over wall-clock time. For a distance increment

$$\Delta s_k = s_k \Delta t_k, \tag{8}$$

the crate uses a first-order distance-domain EMA:

$$\alpha_k = \frac{\Delta s_k}{L + \Delta s_k}, \qquad E_k = (1 - \alpha_k) E_{k-1} + \alpha_k e_k. \tag{9}$$

This is the same small-state first-order update as the time-domain EMA, but the independent variable is traveled distance. L is the effective road interval length. The default $L = 10$ m makes the estimate close to continuous: roughness rises over locally rough pavement and dissipates quickly after the vehicle returns to clean road, without requiring a rolling buffer.

3.5 Interval Accumulation

Most detectors convert an instantaneous metric m_k into an event interval using enter and exit thresholds:

$$\text{enter if} \quad m_k \geq m_{\text{enter}}, \tag{10}$$

$$\text{stay active while} \quad m_k \geq m_{\text{exit}}, \qquad m_{\text{exit}} < m_{\text{enter}}. \tag{11}$$

During an active interval the detector accumulates:

$$T = \sum_k \Delta t_k, \tag{12}$$

$$\bar{m} = \frac{1}{T} \sum_k m_k \Delta t_k, \tag{13}$$

$$m_{\max} = \max_k m_k, \tag{14}$$

$$\bar{s} = \frac{1}{T} \sum_k s_k \Delta t_k. \tag{15}$$

The interval is emitted only if $T \geq T_{\min}$. A refractory time suppresses duplicate events immediately after a valid event.

4 Road Roughness Analyzer

4.1 Metric

`RoadRoughnessAnalyzer` estimates continuous road-surface roughness rather than a discrete event. The input is vehicle-frame, gravity-compensated vertical acceleration. The implementation first rejects grade, body attitude drift, and high-frequency sensor/structure noise:

$$a_{z,\text{hp},k} = \text{HPF}(a_{z,k}; f_c = 0.7 \text{ Hz}), \tag{16}$$

$$a_{z,\text{bp},k} = \text{LPF}(a_{z,\text{hp},k}; f_c = 10 \text{ Hz}). \tag{17}$$

Large isolated defects are limited before they enter the roughness energy:

$$\tilde{a}_{z,k} = \text{clamp}(a_{z,\text{bp},k}, -1.5, 1.5). \tag{18}$$

The roughness energy sample is

$$e_k = \tilde{a}_{z,k}^2. \tag{19}$$

When $s_k \geq 2.0$ m/s, the analyzer updates the distance-domain energy estimate with $L = 10$ m:

$$E_k = (1 - \alpha_k)E_{k-1} + \alpha_k e_k, \qquad \alpha_k = \frac{s_k \Delta t_k}{10 + s_k \Delta t_k}. \tag{20}$$

When speed is below the gate, filter state is kept current but roughness energy and valid roughness distance are held. This prevents stationary vibration and parking-lot maneuvering from being interpreted as road texture.

The reported roughness is

$$R_k = \sqrt{E_k}. \tag{21}$$

Initial qualitative bands are:

Class	Condition	Interpretation
Very smooth	$R < 0.15$ m/s^2	Minimal vertical vibration.
Smooth	$0.15 \leq R < 0.25$ m/s^2	Low vertical vibration.
Light texture	$0.25 \leq R < 0.40$ m/s^2	Mild road texture.
Moderate	$0.40 \leq R < 0.60$ m/s^2	Noticeable sustained texture.
Rough	$0.60 \leq R < 0.90$ m/s^2	Sustained rough pavement.
Very rough	$0.90 \leq R < 1.20$ m/s^2	Strong sustained vibration.
Severe	$R \geq 1.20$ m/s^2	Extreme sustained vibration.

4.2 Why Not Peak Acceleration

Peak vertical acceleration is already used by speed-bump detection. Roughness is intended to describe a road interval, so a single bump or pothole should not dominate the estimate. The band-pass, clipping, speed gate, and 10 m distance-domain EMA make the metric local and streaming while keeping memory use constant.

5 Speed-Bump Detector

5.1 Physical Signature

A road speed bump creates a short vertical impulse sequence as the front axle and rear axle traverse the bump. In gravity-compensated vehicle-frame vertical acceleration, the idealized pattern is an alternating sequence of three extrema:

$$a_{z,\mathrm{hp}}(t_1), \quad a_{z,\mathrm{hp}}(t_2), \quad a_{z,\mathrm{hp}}(t_3), \qquad \mathrm{sgn}(a_1) = \mathrm{sgn}(a_3) \neq \mathrm{sgn}(a_2). \tag{22}$$

Pitch motion is used as corroboration rather than as the primary trigger, because pitch angle alone can also respond to braking, turning on uneven roads, or grade changes.

5.2 Filtered Signals

The detector computes:

$$\theta_{\mathrm{hp},k} = \mathrm{HPF}(\theta_k; f_c = 0.45 \text{ Hz}), \tag{23}$$

$$a_{\mathrm{hp},k} = \mathrm{HPF}(a_{z,k}; f_c = 0.70 \text{ Hz}). \tag{24}$$

Adaptive noise floors are:

$$\sigma_{\theta,k} = \mathrm{EMA}(|\theta_{\mathrm{hp},k}|\,; \tau = 6 \text{ s}), \tag{25}$$
$$\sigma_{a,k} = \mathrm{EMA}(|a_{\mathrm{hp},k}|\,; \tau = 6 \text{ s}). \tag{26}$$

The peak thresholds are:

$$A_{\mathrm{thr}} = \max(3.5\,\sigma_a,\ 1.5 \text{ m/s}^2), \tag{27}$$
$$\Theta_{\mathrm{thr}} = \max(3.0\,\sigma_\theta,\ 0.25^\circ). \tag{28}$$

5.3 Extremum Extraction

The vertical acceleration slope is

$$\dot{a}_{\mathrm{hp},k} = \frac{a_{\mathrm{hp},k} - a_{\mathrm{hp},k-1}}{\Delta t_k}. \tag{29}$$

A local maximum is detected when $\dot{a}_{k-1} > 0$ and $\dot{a}_k \leq 0$; a local minimum is detected when $\dot{a}_{k-1} < 0$ and $\dot{a}_k \geq 0$. An extremum is kept only when its absolute vertical acceleration exceeds the physical floor 1.5 m/s^2. The detector stores only the latest four extrema, so memory use remains constant.

For each extremum interval, the detector also stores the maximum pitch high-pass magnitude since the previous extremum:

$$\Theta_i = \max_{t \in (t_{i-1}, t_i]} |\theta_{\mathrm{hp}}(t)|\,. \tag{30}$$

5.4 Candidate Timing

Each latest-three-extrema group is scored only if the polarities alternate. Let

$$D = t_3 - t_1, \qquad \bar{s} = \frac{s_1 + s_2 + s_3}{3}. \tag{31}$$

The candidate must pass both absolute and speed-adaptive duration bounds:

$$D_{\min} = \max\left(0.18,\ 0.35\frac{L_{\min}}{\bar{s}}\right), \tag{32}$$
$$D_{\max} = \min\left(1.8,\ 2.5\frac{L_{\max}}{\bar{s}}\right), \tag{33}$$

where $L_{\min} = 1.8$ m and $L_{\max} = 3.6$ m are plausible front/rear axle spacing bounds. The speed must also satisfy $\bar{s} \geq 1.5$ m/s.

The detector tracks the total time within the pattern for which $|a_{\mathrm{hp}}| \geq 1.5$ m/s^2. It requires:

$$T_{\mathrm{active}} \geq 0.25 \text{ s}, \tag{34}$$
$$\frac{T_{\mathrm{active}}}{D} \geq 0.25. \tag{35}$$

5.5 Score and Confidence

For extrema magnitudes $A_i = |a_i|$, pitch peaks Θ_i, and candidate duration D, the component scores are:

$$A_{\max} = \max_i A_i, \tag{36}$$

$$\Theta_{\max} = \max_i \Theta_i, \tag{37}$$

$$S_a = \operatorname{clamp}\left(\frac{A_{\max}}{A_{\text{thr}}} - 1,\ 0,\ 1\right), \tag{38}$$

$$S_\theta = \operatorname{clamp}\left(\frac{\Theta_{\max}}{\Theta_{\text{thr}}} - 1,\ 0,\ 1\right), \tag{39}$$

$$S_b = \frac{\min(A_1, A_3)}{\max(A_1, A_3, 10^{-3})}. \tag{40}$$

The spacing score favors the center of the admissible duration interval:

$$D_c = \frac{D_{\min} + D_{\max}}{2}, \qquad W = \frac{\max(D_{\max} - D_{\min}, 10^{-3})}{2}, \tag{41}$$

$$S_D = \operatorname{clamp}\left(1 - \frac{|D - D_c|}{W},\ 0,\ 1\right). \tag{42}$$

The final pattern score is:

$$S = \operatorname{clamp}\left((0.45 S_a + 0.25 S_D + 0.15 S_b + 0.15 S_\theta) S_\theta,\ 0,\ 1\right). \tag{43}$$

The pitch score multiplies the weighted score, so vertical acceleration alone does not emit an event.

An event is emitted when $S \geq 0.12$ and at least four seconds have elapsed since the last bump event. The UI-facing confidence is not a calibrated probability; it is normalized only after the trigger threshold is cleared:

$$c_{\text{display}} = 0.90 + 0.08 \operatorname{clamp}\left(\frac{S - 0.12}{1 - 0.12}, 0, 1\right). \tag{44}$$

6 Hill Detector

The hill detector identifies sustained grade from vehicle pitch. It is a piecewise-constant interval detector with sign:

$$\operatorname{kind}(\theta) = \begin{cases} \text{uphill}, & \theta \geq 4^\circ, \\ \text{downhill}, & \theta \leq -4^\circ, \\ \varnothing, & \text{otherwise.} \end{cases} \tag{45}$$

An interval is emitted when the same signed condition remains active for at least three seconds. During the interval, the detector reports:

$$T = t_{\text{end}} - t_{\text{start}}, \tag{46}$$

$$\bar{\theta} = \frac{1}{T} \int_{t_{\text{start}}}^{t_{\text{end}}} \theta(t)\, dt, \tag{47}$$

$$\theta_{\max} = \max_t |\theta(t)|, \tag{48}$$

$$\bar{s} = \frac{1}{T} \int_{t_{\text{start}}}^{t_{\text{end}}} \max(s(t), 0)\, dt. \tag{49}$$

The implementation emits the event when pitch exits the threshold band or when the detector is explicitly finished at the end of a replay.

7 Reverse Detector

Reverse detection uses vehicle-frame forward velocity directly. The detector starts a candidate when:

$$v_x < -0.5 \text{ m/s}. \tag{50}$$

The candidate is confirmed only after this condition persists for 0.5 s. Once confirmed, the interval remains active while:

$$v_x < -0.2 \text{ m/s}. \tag{51}$$

This hysteresis prevents repeated short events around zero speed. The event is closed after 0.5 s above the exit threshold and emitted if its duration is at least one second.

Reported reverse speed uses positive magnitude:

$$s_{\text{rev}} = \max(-v_x, 0). \tag{52}$$

Mean and peak reverse speeds are accumulated over the confirmed interval.

8 Harsh Acceleration and Braking

8.1 Velocity-Derivative Metric

Harsh longitudinal events are based on changes in forward velocity rather than raw accelerometer samples. This makes the detector less sensitive to road vibration and IMU mounting artifacts. The raw derivative is:

$$a_{x,\text{raw},k} = \text{clamp}\left(\frac{v_{x,k} - v_{x,k-1}}{\Delta t_k},\ -15,\ 15\right)\ \text{m/s}^2. \tag{53}$$

The metric used by the interval detector is an EMA-smoothed derivative:

$$\hat{a}_{x,k} = \text{EMA}(\hat{a}_{x,k-1}, a_{x,\text{raw},k}, \tau = 0.6 \text{ s}, \Delta t_k). \tag{54}$$

The speed gate uses:

$$s_k = \max(|v_{x,k}|, |v_{x,k-1}|). \tag{55}$$

8.2 Harsh Acceleration

If $s_k < 1$ m/s, the metric is forced to zero. Otherwise:

$$m_{\text{accel},k} = \max(\hat{a}_{x,k}, 0). \tag{56}$$

The interval thresholds are:

$$m_{\text{enter}} = 2.5 \text{ m/s}^2, \tag{57}$$
$$m_{\text{exit}} = 2.0 \text{ m/s}^2, \tag{58}$$
$$T_{\text{min}} = 0.4 \text{ s}, \tag{59}$$
$$T_{\text{refractory}} = 2.0 \text{ s}. \tag{60}$$

The emitted event reports duration, mean and peak acceleration metric, mean and peak speed, and net velocity change:

$$\Delta v_x = v_{x,\text{end}} - v_{x,\text{start}}. \tag{61}$$

8.3 Harsh Braking

Braking uses the same smoothed derivative but flips the sign:

$$m_{\text{brake},k} = \max(-\hat{a}_{x,k}, 0). \tag{62}$$

The default thresholds are:

$$m_{\text{enter}} = 3.0 \text{ m/s}^2, \tag{63}$$
$$m_{\text{exit}} = 2.4 \text{ m/s}^2, \tag{64}$$
$$T_{\text{min}} = 0.4 \text{ s}, \tag{65}$$
$$T_{\text{refractory}} = 2.0 \text{ s}. \tag{66}$$

The reported Δv_x is signed, so braking events normally have $\Delta v_x < 0$.

9 Harsh Cornering

Harsh cornering uses the steady-turn kinematic relation:

$$a_y \approx v\omega_z. \tag{67}$$

The detector metric is:

$$m_{\text{corner},k} = \begin{cases} |s_k \omega_{z,k}|, & s_k \geq 3 \text{ m/s}, \\ 0, & s_k < 3 \text{ m/s}. \end{cases} \tag{68}$$

The default interval thresholds are:

$$m_{\text{enter}} = 3.0 \text{ m/s}^2, \tag{69}$$
$$m_{\text{exit}} = 2.4 \text{ m/s}^2, \tag{70}$$
$$T_{\text{min}} = 0.5 \text{ s}, \tag{71}$$
$$T_{\text{refractory}} = 2.0 \text{ s}. \tag{72}$$

The steady-turn approximation is deliberately simple. It is robust for embedded event detection because it only needs speed and yaw rate, and it does not depend on lateral accelerometer calibration or road bank compensation.

10 Default Configuration Summary

Detector	**Default**	**Purpose**
Speed bump pitch HPF	0.45 Hz	Reject grade and slow pitch.
Speed bump vertical accel HPF	0.70 Hz	Reject slow body motion.
Speed bump noise EMA	6.0 s	Adaptive vibration floor.
Speed bump minimum speed	1.5 m/s	Avoid static/creep false positives.
Speed bump duration	0.18–1.8 s	Absolute event timing.

Speed bump wheelbase bounds	1.8–3.6 m	Speed-adaptive timing.
Hill pitch threshold	4°	Sustained uphill/downhill grade.
Hill minimum duration	3.0 s	Suppress short pitch transients.
Reverse enter/exit	$-0.5/-0.2$ m/s	Hysteresis around zero speed.
Reverse minimum duration	1.0 s	Suppress brief backing corrections.
Harsh accel enter/exit	2.5/2.0 m/s^2	Longitudinal acceleration hysteresis.
Harsh brake enter/exit	3.0/2.4 m/s^2	Longitudinal deceleration hysteresis.
Harsh corner enter/exit	3.0/2.4 m/s^2	Lateral acceleration hysteresis.

11 Trip Statistics

`TripStats` is a constant-memory accumulator that runs beside the event detectors. It is not an offline reducer: each call processes one `TripSample`, updates a fixed set of sums and extrema, and returns no allocation-backed history. The main integrated quantities are:

$$D = \sum_k \bar{s}_k \Delta t_k, \tag{73}$$

$$T_{\text{moving}} = \sum_k \mathbb{1}(\bar{s}_k \geq s_{\text{move}}) \Delta t_k, \tag{74}$$

$$D_{\text{rev}} = \sum_k \max(-\bar{v}_{x,k}, 0) \Delta t_k. \tag{75}$$

Here $\bar{s}_k$ and $\bar{v}_{x,k}$ are one-sample trapezoidal averages over the current interval. The trapezoid uses only the previous sample, so it remains embedded-friendly while avoiding the step-change bias from assigning a whole interval to the newest sample.

When a caller supplies a vertical position h_k in the sample stream, the accumulator computes elevation gain and loss:

$$\Delta h_k = h_k - h_{k-1}. \tag{76}$$

Positive Δh_k contributes to elevation gain and negative Δh_k contributes to elevation loss. In the simulator, $h_k = -D_{\text{GNSS}}$, where D_{GNSS} is the GNSS down coordinate in the fixed first-GNSS NED frame. The trip accumulator ignores height deltas across different `height_frame_id` values, so callers that use a local reanchored frame can avoid false elevation jumps by changing the frame identifier whenever the anchor changes.

Rolling quantities use the same EMA primitive as the detectors:

$$\bar{s}_k^{\text{roll}} = \text{EMA}(\bar{s}_{k-1}^{\text{roll}}, s_k, \tau, \Delta t_k), \tag{77}$$

$$\bar{a}_k^{\text{roll}} = \text{EMA}(\bar{a}_{k-1}^{\text{roll}}, |a_k|, \tau, \Delta t_k). \tag{78}$$

These are approximate recent-behavior summaries, not exact last-N-second windows. Exact windows would require a ring buffer; the trip accumulator intentionally avoids that memory cost.

The accumulator also records detector event counts by category. Normalized rates such as bumps per kilometer, harsh events per kilometer, and reverse seconds per kilometer are computed from the running counters and distance.

12 Visualizer Interpretation

The map markers and hover mini-plots are visualization aids. They do not feed back into the detectors. For each event, the visualizer draws a short local window around the detected interval, currently with five seconds of padding on both sides when data is available:

$$[t_{\text{start}} - 5,\ t_{\text{end}} + 5]. \tag{79}$$

The mini-plot shows the signal that caused the detector to trigger, for example vertical acceleration and pitch for speed bumps, longitudinal acceleration for harsh acceleration/braking, and yaw-rate-derived lateral acceleration for harsh cornering.

13 Design Rationale

The event crate favors deterministic low-memory logic over more expressive offline methods. Windowed FFTs, matched filters, and machine-learning classifiers can be useful for richer offline analysis, but they introduce larger buffers, more tuning surface, and more deployment complexity. The current detectors instead use physically interpretable metrics:

- speed bumps: alternating high-pass vertical acceleration extrema corroborated by pitch motion,
- hills: sustained pitch sign and magnitude,
- reverse: signed vehicle-frame forward velocity with hysteresis,
- harsh acceleration/braking: smoothed velocity derivative,
- harsh cornering: steady-turn lateral acceleration $v\omega_z$.

This keeps the algorithms easy to port to small embedded targets and easy to debug from the visualizer.